

Instituto Tecnológico de Las Américas

ITLA

Laboratorio: Infraestructura DMZ con FortiGate

DMZ, VLANs, políticas de seguridad y Web Filter

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Matrícula: 2024-2356

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LINK DEL VIDEO: [Tarea Semana 4 \(Practica 3\)](#)

1. Propósito del laboratorio

En este laboratorio se montó la red de una empresa con un solo FortiGate que separa a los usuarios de los servidores. Los tres servidores (Sistema de Caja, Sistema de Inventario y Base de Datos) quedaron en una DMZ, y los usuarios en dos VLAN con permisos diferentes: la VLAN 20 puede administrar los servidores por SSH, y la VLAN 10 solo puede entrar al Sistema de Caja. Si la VLAN 10 intenta abrir el Inventario, el FortiGate le muestra una página de bloqueo.

Para montarlo se clonó la Infraestructura 1 del laboratorio de VPN y se reutilizó el FortiGate, el ISP y el switch. Toda la configuración del FortiGate se hizo por la GUI.

2. Topología y direccionamiento

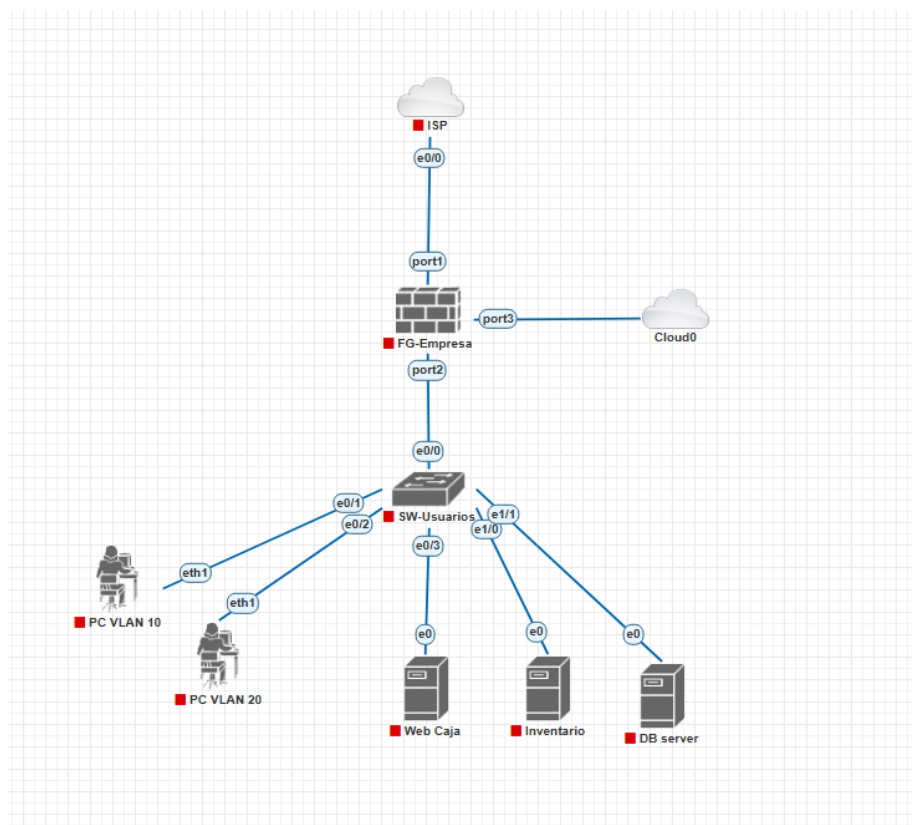


Figura 1. Topología implementada en PNETLab.

En el diagrama lógico se ve cómo queda cada enlace: el FortiGate sale al ISP por el port1, se gestiona por el port3 desde Cloud0, y por el port2 lleva las tres VLAN al switch en un solo trunk.

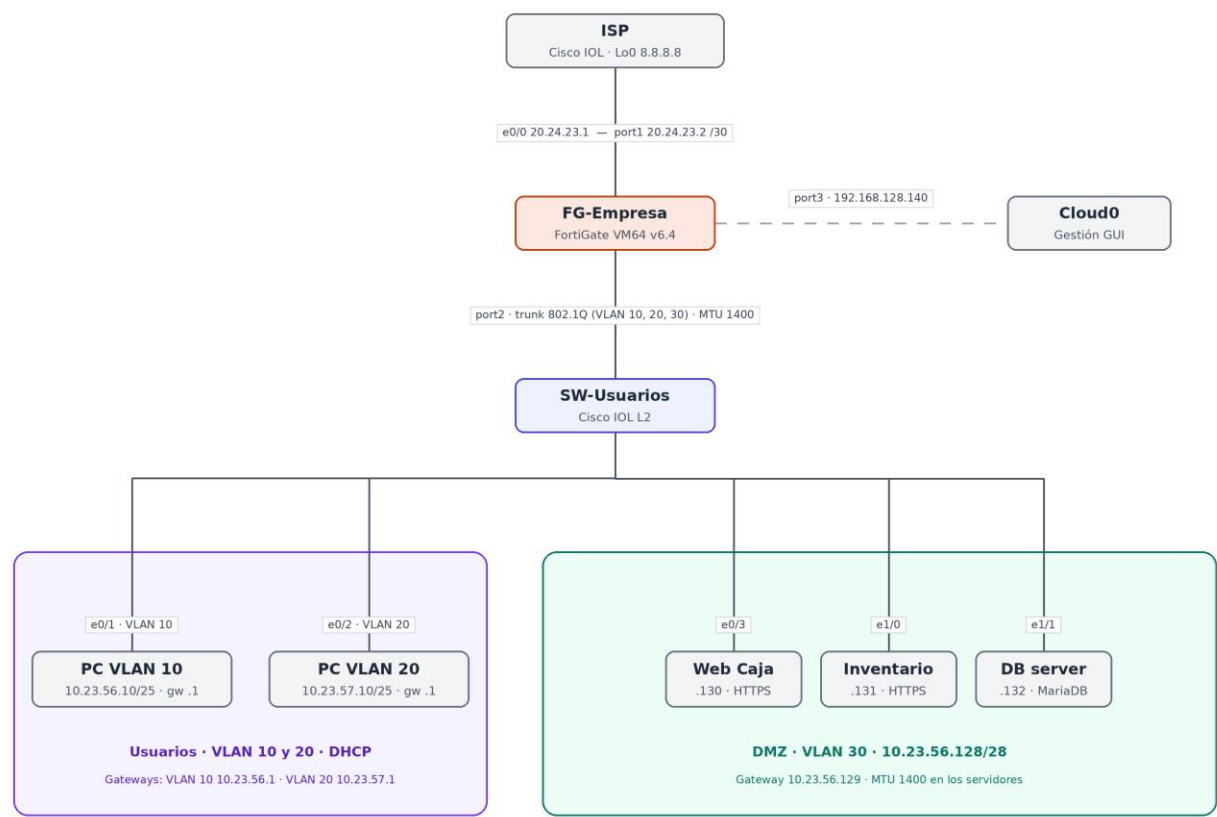


Figura 2. Diagrama lógico con direccionamiento, puertos y VLAN.

El direccionamiento sale de la matrícula 2024-2356. Las dos VLAN de usuarios son /25 y la DMZ es /28, como pide la práctica.

Segmento	Red	Gateway	Notas
WAN	20.24.23.0/30	20.24.23.1 (ISP)	FG-Empresa port1 .2
VLAN 10	10.23.56.0/25	10.23.56.1	DHCP .10 a .100
VLAN 20	10.23.57.0/25	10.23.57.1	DHCP .10 a .100
DMZ (VLAN 30)	10.23.56.128/28	10.23.56.129	Caja .130, Inventario .131, DB .132

3. Switch y seguridad básica

En el switch se crearon las VLAN 10, 20 y 30, el trunk hacia el FortiGate con VLAN nativa 99 y solo las VLAN necesarias. En los puertos de acceso se activó port-security con MAC sticky, BPDU Guard y DHCP snooping. Los puertos que no se usan quedaron apagados en la VLAN 999.

```
SW-Usuarios#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Et0/0	on	802.1q	trunking	99

Port	Vlans allowed on trunk
Et0/0	10,20,30

```
SW-Usuarios#show port-security address
```

Vlan	Mac Address	Type	Ports
10	5000.0040.0001	SecureSticky	Et0/1
20	5000.003d.0001	SecureSticky	Et0/2
30	50c7.f400.3e00	SecureSticky	Et0/3
30	506c.da00.4100	SecureSticky	Et1/0
30	5082.6600.4200	SecureSticky	Et1/1

Cada puerto solo acepta la MAC del equipo que tiene conectado. Los contadores de violación que muestra show port-security son de cuando el switch aprendió MACs temporales de PNETLab, y demuestran que la protección sí actúa.

4. Configuración del FortiGate

4.1 Interfaces VLAN

Sobre el port2 se crearon las tres VLAN. La 10 y la 20 tienen servidor DHCP, y la DMZ no, porque los servidores llevan IP fija.

New Interface

Name

Alias

Type

Interface

VLAN ID

Role

Address

Addressing mode

IP/Netmask

Create address object matching subnet ☒

Name

Destination

Secondary IP address ☐

Administrative access

IPv4 ☐ HTTPS ☒ PING ☐ FMG-Access

☐ SSH ☐ SNMP ☐ FTM

☐ RADIUS Accounting ☐ Security Fabric Connection

☒ DHCP Server

Address range

Netmask

Default gateway

DNS server

Lease time ☒ second(s)

Advanced

Figura 3. VLAN 10 con su IP y servidor DHCP.

New Interface

Name: VLAN20
Alias: USUARIOS-V20
Type: VLAN
Interface: port2
VLAN ID: 20
Role: LAN

Address

Addressing mode: Manual DHCP
IP/Netmask: 10.23.57.1/25
Create address object matching subnet: ☒
Name: VLAN20 address
Destination: 10.23.57.1/25
Secondary IP address: ☐

Administrative access

IPv4: ☐ HTTPS ☒ PING ☐ FMG-Access
☐ SSH ☐ SNMP ☐ FTM
☐ RADIUS Accounting ☐ Security Fabric Connection

☒ DHCP Server

Address range: 10.23.57.10-10.23.57.100
Netmask: 255.255.255.128
Default gateway: Same as Interface IP Specify
DNS server: Same as System DNS Same as Interface IP Specify 8.8.8.8
Lease time: ☒ 604800 second(s)

☒ Advanced

OK Cancel

Figura 4. VLAN 20 con su IP y servidor DHCP.

New Interface

Name: DMZ
 Alias: DMZ
 Type: VLAN
 Interface: port2
 VLAN ID: 30
 Role: DMZ

Address
 Addressing mode: Manual | DHCP
 IP/Netmask: 10.23.56.129/28
 Create address object matching subnet: ☒
 Name: DMZ address
 Destination: 10.23.56.129/28
 Secondary IP address: ☐

Administrative access
 IPv4: ☐ HTTPS ☒ PING ☐ FMG-Access
☐ SSH ☐ SNMP ☐ FTM
☐ RADIUS Accounting ☐ Security Fabric Connection

Network
 Device detection: ☐

Traffic Shaping
 Outbound shaping profile: ☐

Miscellaneous
 Comments: 0/255
 Status: ☒ Enabled ☐ Disabled

OK Cancel

Figura 5. Interfaz DMZ (VLAN 30) con rol DMZ.

4.2 Políticas

La licencia de evaluación solo permite 5 políticas, así que algunas se combinaron. Todo lo que no está permitido lo bloquea la Implicit Deny, que tiene el log activado.

Política	Origen → Destino	Servicio	Detalle
Usuarios-a-Internet	VLAN10, VLAN20 → WAN	ALL	Con NAT
V10-a-Inventario	VLAN10 → Inventario	HTTP	Web Filter: página de bloqueo
V20-a-DMZ	VLAN20 → DMZ	HTTPS, SSH	Única con SSH
V10-a-Caja	VLAN10 → Caja	HTTPS	Sin NAT
Implicit Deny	any → any	ALL	Bloquea lo demás, con log

Figura 6. Lista final de políticas del FortiGate.

Edit Policy

Name	Usuarios-a-Internet	
Incoming Interface	USUARIOS-V10 (VLAN10) ✕ USUARIOS-V20 (VLAN20) ✕ +	
Outgoing Interface	WAN (port1) ✕ +	
Source	VLAN10 address ✕ VLAN20 address ✕ +	
Destination	all ✕ +	
Schedule	always	
Service	ALL ✕ +	
Action	✓ ACCEPT ✗ DENY	
Inspection Mode	Flow-based Proxy-based	

Firewall / Network Options

NAT ☒

IP Pool Configuration

Use Outgoing Interface Address

Use Dynamic IP Pool

Preserve Source Port ☐

Protocol Options

PIX default

Security Profiles

AntiVirus ☐

Web Filter ☐

DNS Filter ☐

Application Control ☐

IPS ☐

SSL Inspection

SSL no-inspection

OK Cancel

Figura 7. Política de salida a Internet para las dos VLAN.

FG-Empresa

New Policy

Name: V10-a-Caja

Incoming Interface: USUARIOS-V10 (VLAN10)

Outgoing Interface: DMZ (DMZ)

Source: VLAN10 address

Destination: Web-Caja

Schedule: always

Service: HTTPS

Action: ☒ ACCEPT ☐ DENY

Inspection Mode: ☒ Flow-based ☐ Proxy-based

Firewall / Network Options

NAT: ☐

Protocol Options: ☒ default

Security Profiles

AntiVirus: ☐

Web Filter: ☐

DNS Filter: ☐

Application Control: ☐

IPS: ☐

SSL Inspection: ☒ no-inspection

Logging Options

Log Allowed Traffic: ☒ Security Events ☒ All Sessions

Generate Logs when Session Starts: ☐

OK Cancel

Figura 9. Política V10-a-Caja, solo HTTPS al Sistema de Caja.

4.3 Web Filter

El perfil Bloqueo-Inventario tiene un filtro de URL estático que bloquea 10.23.56.131. Se aplica en la política V10-a-Inventario, en modo proxy, para que el FortiGate responda con su propia página de bloqueo.

Edit Web Filter Profile

Name: Bloqueo-Inventario

Comments: Write a comment... 0/255

Feature set: ☐ Flow-based ☒ Proxy-based

☐ FortiGuard category based filter

☐ Allow users to override blocked categories

☒ File Filter

Log: ☒

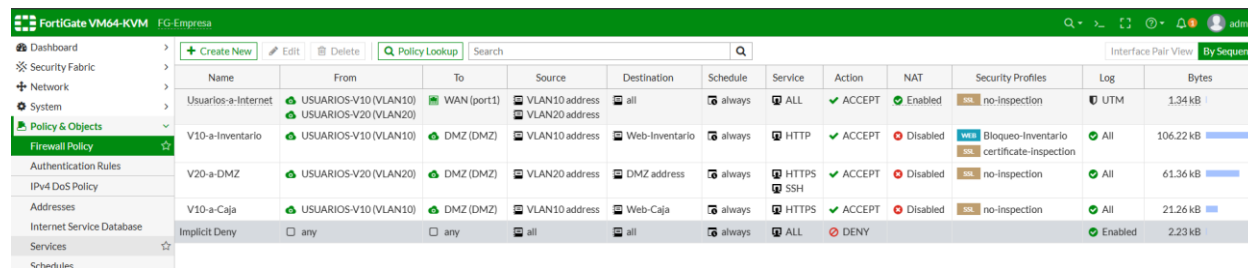
Scan Archived Contents: ☒

[+ Create New](#) [Edit](#) [Delete](#)

Name	Comments	Action	Direction	Protocol	File Types	Match Password-Pr
No results						

Figura 10. Perfil de Web Filter Bloqueo-Inventario.

4.4 Salida de la DMZ a Internet



Name	From	To	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
Usuarios-a-Internet	USUARIOS-V10 (VLAN10)	WAN (port1)	VLAN10 address	all	always	ALL	ACCEPT	Enabled	no-inspection	UTM	1.34 KB
V10-a-Inventario	USUARIOS-V10 (VLAN10)	DMZ (DMZ)	VLAN10 address	Web-Inventario	always	HTTP	ACCEPT	Disabled	Bloqueo-Inventario	All	106.22 kB
V20-a-DMZ	USUARIOS-V20 (VLAN20)	DMZ (DMZ)	VLAN20 address	DMZ address	always	HTTPS	ACCEPT	Disabled	no-inspection	All	61.36 kB
V10-a-Caja	USUARIOS-V10 (VLAN10)	DMZ (DMZ)	VLAN10 address	Web-Caja	always	HTTPS	ACCEPT	Disabled	no-inspection	All	21.26 kB
Implicit Deny	any	any	all	all	always	ALL	DENY			Enabled	2.23 kB

Figura 11. Política de actualizaciones de la DMZ (pendiente).

5. Servidores

Los tres servidores son Ubuntu Server 20.04 con IP fija en la DMZ. Caja e Inventario tienen Apache con HTTPS, y el DB tiene MariaDB con las bases caja e inventario. Los tres tienen SSH con el inicio de sesión de root desactivado.

```
Last login: Tue Oct 6 03:06:32 CST 2026 from 192.168.128.1 on pts/0
root@kum:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 50:c7:f4:00:3e:00 brd ff:ff:ff:ff:ff:ff
    inet 10.23.56.130/28 brd 10.23.56.143 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::52c7:f4ff:fe00:3e00/64 scope link
        valid_lft forever preferred_lft forever
root@kum:~#
```

Figura 12. Web Caja con la IP 10.23.56.130/28.

```
Last login: Tue Oct 6 03:22:03 CST 2026 from 192.168.128.1 on pts/0
root@kum:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 50:6c:da:00:41:00 brd ff:ff:ff:ff:ff:ff
    inet 10.23.56.131/28 brd 10.23.56.143 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::526c:daff:fe00:4100/64 scope link
        valid_lft forever preferred_lft forever
root@kum:~#
```

Figura 13. Inventario con la IP 10.23.56.131/28.

```

root@kvm:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 50:82:66:00:42:00 brd ff:ff:ff:ff:ff:ff
    inet 10.23.56.132/28 brd 10.23.56.143 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::5282:66ff:fe00:4200/64 scope link
        valid_lft forever preferred_lft forever
root@kvm:~# mysql -e "SHOW DATABASES;"
+-----+
| Database |
+-----+
| ca.ja    |
| information_schema |
| inventario |
| mysql    |
| performance_schema |
+-----+
root@kvm:~# systemctl is-active ssh mariadb
active
active
root@kvm:~# ping -c 4 10.23.56.129
PING 10.23.56.129 (10.23.56.129) 56(84) bytes of data.
64 bytes from 10.23.56.129: icmp_seq=1 ttl=255 time=0.657 ms
64 bytes from 10.23.56.129: icmp_seq=2 ttl=255 time=0.674 ms
64 bytes from 10.23.56.129: icmp_seq=3 ttl=255 time=0.703 ms
64 bytes from 10.23.56.129: icmp_seq=4 ttl=255 time=0.764 ms

--- 10.23.56.129 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3020ms
rtt min/avg/max/mdev = 0.657/0.699/0.764/0.040 ms
root@kvm:~#

```

Figura 14. DB server con sus bases de datos, servicios activos y ping al gateway.

6. Pruebas

6.1 DHCP en las VLAN

```
root@PC VLAN 10:/home# dhclient eth1
cmp: EOF on /tmp/tmp.JeQz10LjGh which is empty
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
root@PC VLAN 10:/home# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
54: eth0@if55: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP
    group default
    link/ether 02:42:0a:b1:00:03 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.177.0.3/16 brd 10.177.255.255 scope global eth0
        valid_lft forever preferred_lft forever
57: docker64_0@if56: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group
    default qlen 1000
    link/ether 50:00:00:40:00:00 brd ff:ff:ff:ff:ff:ff link-netnsid 0
59: eth1@if58: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP
    group default qlen 1000
    link/ether 50:00:00:40:00:01 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.23.56.10/25 brd 10.23.56.127 scope global dynamic eth1
        valid_lft 604800sec preferred_lft 604800sec
root@PC VLAN 10:/home# ip route
default via 10.23.56.1 dev eth1
10.23.56.0/25 dev eth1 proto kernel scope link src 10.23.56.10
10.177.0.0/16 dev eth0 proto kernel scope link src 10.177.0.3
root@PC VLAN 10:/home# ping -c 4 10.23.56.1
PING 10.23.56.1 (10.23.56.1) 56(84) bytes of data.
64 bytes from 10.23.56.1: icmp_seq=1 ttl=255 time=0.870 ms
64 bytes from 10.23.56.1: icmp_seq=2 ttl=255 time=0.504 ms
64 bytes from 10.23.56.1: icmp_seq=3 ttl=255 time=0.568 ms
64 bytes from 10.23.56.1: icmp_seq=4 ttl=255 time=0.550 ms

--- 10.23.56.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3055ms
rtt min/avg/max/mdev = 0.504/0.623/0.870/0.144 ms
root@PC VLAN 10:/home# ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=254 time=0.864 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=254 time=1.97 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=254 time=0.790 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=254 time=4.76 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3049ms
rtt min/avg/max/mdev = 0.790/2.097/4.761/1.607 ms
root@PC VLAN 10:/home# ^C
root@PC VLAN 10:/home#
```

Figura 15. PC VLAN 10 recibe 10.23.56.10 y llega al gateway y al ISP.

```

root@PC VLAN 20:/home# ip a show eth1
39: eth1@if38: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP
    group default qlen 1000
    link/ether 50:00:00:3d:00:01 brd ff:ff:ff:ff:ff:ff link-netnsid 0
root@PC VLAN 20:/home# ip link set eth1 up
root@PC VLAN 20:/home# ^C
root@PC VLAN 20:/home# dhclient eth1
cmp: EOF on /tmp/tmp.sySKs3lqMb which is empty
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
root@PC VLAN 20:/home# ^C
root@PC VLAN 20:/home# ip a show eth1
39: eth1@if38: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP
    group default qlen 1000
    link/ether 50:00:00:3d:00:01 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.23.57.10/25 brd 10.23.57.127 scope global dynamic eth1
        valid_lft 604746sec preferred_lft 604746sec
root@PC VLAN 20:/home# ip route
default via 10.23.57.1 dev eth1
10.23.57.0/25 dev eth1 proto kernel scope link src 10.23.57.10
10.177.0.0/16 dev eth0 proto kernel scope link src 10.177.0.2
root@PC VLAN 20:/home# ping -c 4 10.23.57.1
PING 10.23.57.1 (10.23.57.1) 56(84) bytes of data:
64 bytes from 10.23.57.1: icmp_seq=1 ttl=255 time=0.678 ms
64 bytes from 10.23.57.1: icmp_seq=2 ttl=255 time=0.529 ms
64 bytes from 10.23.57.1: icmp_seq=3 ttl=255 time=0.577 ms
64 bytes from 10.23.57.1: icmp_seq=4 ttl=255 time=0.481 ms

--- 10.23.57.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3051ms
rtt min/avg/max/mdev = 0.481/0.566/0.678/0.072 ms
root@PC VLAN 20:/home# ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=254 time=0.962 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=254 time=0.761 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=254 time=0.911 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=254 time=0.850 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3073ms
rtt min/avg/max/mdev = 0.761/0.871/0.962/0.074 ms
root@PC VLAN 20:/home#

```

Figura 16. PC VLAN 20 recibe 10.23.57.10 y llega al gateway y al ISP.

6.2 Acceso desde la VLAN 10

La VLAN 10 entra al Sistema de Caja. El Inventario por HTTPS y el SSH no conectan, porque no hay ninguna política que los permita.

```

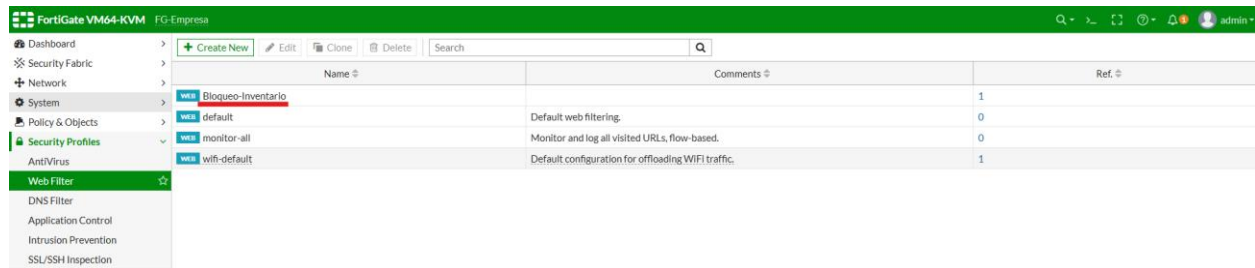
root@PC VLAN 10:/home# curl -k https://10.23.56.130
<!DOCTYPE html>
<html lang="es">
<head><meta charset="UTF-8"><title>Sistema de Caja</title></head>
<body style="font-family: Arial; text-align: center; margin-top: 80px;">
  <h1>Sistema de Caja</h1>
  <p>Laboratorio DMZ - Angel Alcantara - Matricula 2024-2356 - ITLA</p>
  <p>IP del servidor: 10.23.56.130</p>
</body>
</html>
root@PC VLAN 10:/home# curl -k --max-time 5 https://10.23.56.131
curl: (28) Connection timed out after 5001 milliseconds
root@PC VLAN 10:/home# ssh -o ConnectTimeout=5 angel@10.23.56.130
ssh: connect to host 10.23.56.130 port 22: Connection timed out
root@PC VLAN 10:/home#

```

Figura 17. Pruebas desde el PC VLAN 10: Caja carga, Inventario y SSH no.

Cuando intenta entrar al Inventario por HTTP, el FortiGate le responde con la página de bloqueo:

```
root@PC VLAN 10:/home# curl http://10.23.56.131
...
<title>The URL you requested has been blocked</title>
...
<h3>Web Page Blocked!</h3>
<p>The page you have requested has been blocked, because the URL is banned.</p>
URL: http://10.23.56.131/
URL Source: Local URLfilter Block
```



Name	Comments	Ref
web Bloqueo-Inventario		1
web default	Default web filtering.	0
web monitor-all	Monitor and log all visited URLs, flow-based.	0
web wifi-default	Default configuration for offloading WIFI traffic.	1

Figura 18. Log del bloqueo en Log & Report > Web Filter.

6.3 Acceso desde la VLAN 20

La VLAN 20 entra a las dos páginas y también por SSH. Es la única VLAN con SSH hacia los servidores.

```

root@PC VLAN 20:/home# curl -k https://10.23.56.130
<!DOCTYPE html>
<html lang="es">
<head><meta charset="UTF-8"><title>Sistema de Caja</title></head>
<body style="font-family: Arial; text-align: center; margin-top: 80px;">
  <h1>Sistema de Caja</h1>
  <p>Laboratorio DMZ - Angel Alcantara - Matricula 2024-2356 - ITLA</p>
  <p>IP del servidor: 10.23.56.130</p>
</body>
</html>
root@PC VLAN 20:/home# curl -k --max-time 5 https://10.23.56.131
<!DOCTYPE html>
<html lang="es">
<head><meta charset="UTF-8"><title>Sistema de Inventario</title></head>
<body style="font-family: Arial; text-align: center; margin-top: 80px;">
  <h1>Sistema de Inventario</h1>
  <p>Laboratorio DMZ - Angel Alcantara - Matricula 2024-2356 - ITLA</p>
  <p>IP del servidor: 10.23.56.131</p>
</body>
</html>
root@PC VLAN 20:/home# ssh -o ConnectTimeout=5 angel@10.23.56.130
angel@10.23.56.130's password:
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-29-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Tue 06 Oct 2026 04:09:12 AM CST

System load:  0.0          Processes:      115
Usage of /:   2.3% of 96.46GB Users logged in: 1
Memory usage: 8%          IPv4 address for eth0: 10.23.56.130
Swap usage:  0%

345 updates can be installed immediately.
247 of these updates are security updates.
To see these additional updates run: apt list --upgradable

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Tue Oct  6 04:05:19 2026 from 10.23.57.10
angel@kvm:~$ █

```

Figura 19. Pruebas desde el PC VLAN 20: Caja, Inventario y SSH funcionan.

6.4 Fuga de tráfico desde la DMZ

Desde el DB server se intentó llegar a los PCs de las dos VLAN, y los dos pings fallaron. Si un servidor de la DMZ es comprometido, no se puede usar para atacar a los usuarios.

```

angel@kvm:~$ ping -c 4 10.23.56.10
4 packets transmitted, 0 received, 100% packet loss, time 3076ms

angel@kvm:~$ ping -c 4 10.23.57.10
4 packets transmitted, 0 received, 100% packet loss, time 3069ms

```


6.5 Traceroute

```
root@PC VLAN 10:/home# traceroute -n -T -p 443 10.23.56.130
1 10.23.56.1 7.802 ms
2 10.23.56.130 4.021 ms

root@PC VLAN 20:/home# traceroute -n -T -p 443 10.23.56.130
1 10.23.57.1 1.550 ms
2 10.23.56.130 1.519 ms
```

Se usó traceroute TCP al puerto 443 porque las políticas solo permiten HTTPS. El servidor aparece a dos saltos: el gateway de cada VLAN y el servidor.

7. Problemas encontrados

Port-security y PNETLab: el switch aprendía como sticky una MAC temporal que PNETLab usa al encender los nodos, y bloqueaba al equipo real. Se resolvió reiniciando el port-security en esos puertos.

MTU en el trunk: el HTTPS se quedaba en el Client hello porque los paquetes de 1500 bytes se perdían en el trunk por la etiqueta 802.1Q. Se bajó el MTU a 1400 en los servidores y en el port2 del FortiGate.

```

root@PC_VLAN_20:/home# ssh angel@10.23.56.130
The authenticity of host '10.23.56.130 (10.23.56.130)' can't be established.
RSA key fingerprint is SHA256:k7FKuqXeczSGlf8Ye6vlgc/Rwdu72e39RQSBfk9jyM.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.23.56.130' (RSA) to the list of known hosts.
angel@10.23.56.130's password:
Welcome to Ubuntu 20.04 LTS (GNU/Linux 5.4.0-29-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

System information as of Tue 06 Oct 2026 04:05:19 AM CST

System load:  0.0          Processes:      114
Usage of /:   2.3% of 96.46GB    Users logged in: 1
Memory usage: 8%              IPv4 address for eth0: 10.23.56.130
Swap usage:  0%

345 updates can be installed immediately.
247 of these updates are security updates.
To see these additional updates run: apt list --upgradable

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

angel@kvm:~$ su -
Password:
root@kvm:~# sed -i '/mtu: 1400/d' /etc/netplan/01-lab.yaml
root@kvm:~# sed -i 's/^(\s*)addresses: \[.*\]/\1addresses: \[2\n\1mtu: 1400\]/' /
etc/netplan/01-lab.yaml
root@kvm:~# cat /etc/netplan/01-lab.yaml
network:
  version: 2
  ethernet:
    eth0:
      addresses: [10.23.56.130/28]
      mtu: 1400
      routes:
        - to: 0.0.0.0/0
          via: 10.23.56.129
root@kvm:~# netplan apply
root@kvm:~# ip link show eth0 | grep -o "mtu [0-9]*"
mtu 1400
root@kvm:~# exit
logout
angel@kvm:~$ exit
logout
Connection to 10.23.56.130 closed.
root@PC_VLAN_20:/home#

```

Figura 20. Corrección del MTU en el servidor desde la VLAN 20 por SSH.

Límite de políticas: la licencia de evaluación no deja crear más de 5 políticas, así que se combinaron reglas sin perder ningún requisito.

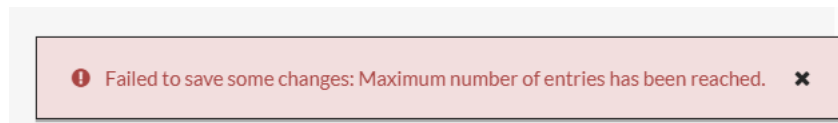


Figura 21. Error de la licencia al crear la sexta política.

Deep inspection: con la licencia de evaluación el FortiGate no descifra el HTTPS, por eso la página de bloqueo se muestra sobre HTTP.

8. Conclusión

Se logró separar a los usuarios de los servidores con una DMZ, controlar qué puede hacer cada VLAN y evitar que la DMZ llegue a la LAN. Lo más interesante fue ver la diferencia entre bloquear con una política, que solo corta la conexión, y bloquear con el Web Filter, que le muestra al usuario que violó una política. También quedó claro que la licencia de evaluación limita bastante, y que hay que buscar alternativas que cumplan el objetivo.